MIL-STD-1163
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MILITARY STANDARD

LITHOGRAPHIC CHEMICALS

FSC 6850

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DEPARTMENT OF DEFENSE

Washington, D. C. 20301

LITHOGRAPHIC CHEMICALS, MIL-STD-1163

1. This Military Standard is mandatory for use by all Departments and Agencies of the Department of Defense.

2. Recommended corrections, additions or deletions should be addressed to U. S. Army Natick Laboratories (GL), Natick, Massachusetts 01760.

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FOREWORD

This document has been developed to designate the standard items to be used by the Military and Defense Agencies. It provides a basis for procuring items for better performance and care, resulting in longer life of lithographic equipment.

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1. SCOPE

1.1 Purpose.- The purpose of this standard is to designate the standard items for use in Military applications and to provide performance characteristics, technical description, uses and limitations, interchangeability, safety and handling precautions, labeling information, specific packaging, storage data (including shelf life) and applicable procurement documentation for each designated standard item.

1.2 Application.- This standard does not necessarily include all classifications of the items represented by the title or those which are commercially available. Items listed herein accommodate essential requirements of the Military Departments and Defense Agencies, and when these items are properly used in new design and applications, they will effect continued economies in all logistics functions.

1.3 Coverage.- This standard covers the following items.

Name	No. of Items
Fountain solution	2
Lithographic blanket-roller wash	2
Lithographic plate sensitizing coating surface	2
Lithographic solvent	1
Regenerator, lithographic blanket	1
Lithographic plate asphaltum solution	1
Lithographic plate preserving gum	1

2. REFERENCED DOCUMENTS

2.1 The issues of the following documents in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-F-615	-	Fountain Solution, Aluminum or Zinc Plate, for Offset Duplicating Process.
O-L-298	-	Lithographic, Blanket-Roller, Wash (For Daily Use).
O-L-300	-	Lithographic Plate Sensitizing Coating, Surface.
O-L-1031	-	Lithographic Solvent.
O-R-1151	-	Regenerator, Lithographic Blanket.
O-T-236	-	Tetrachloroethylene (Perchloroethylene) Technical Grade.

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P-D-680	-	Dry Cleaning Solvent.
TT-I-558	-	Ink, Marking Stencil, Opaque, for Nonporous Surfaces, Metals, Glass, etc.
TT-L-132	-	Lithographic Plate Asphaltum Solution.
JJJ-L-821	-	Lithographic Plate Preserving Gum.
PPP-C-186	-	Containers, Packaging and Packing, for Drugs, Chemicals, and Pharmaceuticals.

2.2 Other publications.- The following document forms a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

FEDERAL HAZARDOUS SUBSTANCES LABELING ACT

(Application for copies may be made to U. S. Department of Health, Education, and Welfare, Food and Drug Administration, Washington, D. C. 20203.)

3. DEFINITIONS AND ABBREVIATIONS

Av.	-	Avoirdupois.
Be.	-	Baumé. Baumé is a unit for measuring density of liquid.
Boiling point	-	Boiling point of a substance is the temperature at which the vapor pressure of the substance is slightly greater than the opposing pressure.
C.	-	Centigrade.
cc	-	Cubic centimeter.
End point	-	End point is the temperature at which the liquid ceases to distill over.
Flash point	-	The lowest temperature at which a combustible material will give off a flammable vapor which will burn momentarily.
F.	-	Fahrenheit.
gm	-	Grams.
liq.	-	Liquid.
oz.	-	Ounce.
pH	-	pH is considered to be the negative logarithm (to the base 10) of the hydrogen ion concentration of a solution.
Specific gravity	-	Specific gravity is the ratio of the mass of the body to the mass of an equal volume of water at 4°C. or other specified temperature.

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4. GENERAL REQUIREMENTS

4.1 The general requirements comprise the established frequencies listed and modified in 5.1, 5.2, 5.3, 5.4, 5.5, 5.6, and 5.7.

5. DETAIL REQUIREMENTS

5.1 Name.- Fountain solution.

5.1.1 Specification.- O-F-615 - Fountain Solution, Aluminum or Zinc Plate, for Offset Duplicating Process.

5.1.2 Performance.- The fountain solution is a concentrated solution which, when diluted to the desired pH, repels ink in the non-printing areas of the plate.

5.1.3 Technical description.-

5.1.3.1 The fountain solution for aluminum and zinc plates shall have the following composition:

SOLUTION A

Water	1,000 cc	or	32 liq. oz.
Ammonium Bichromate	45 grams	or	1-1/2 avoir oz.
Phosphoric Acid, 85 percent	28.8 cc	or	3/4 liq. oz.

SOLUTION B

Magnesium Nitrate	120 grams	or	16 avoir oz.
Water to make	1,000 cc	or	1 gallon

The press fountain solution shall be as follows:

Solution A	4 cc	or	1/2 liq. oz.
Solution B	8 cc	or	1 liq. oz.
Gum Arabic, 14°be.	8 cc	or	1 liq. oz.
Water	1,000 cc	or	1 gallon

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The fountain solution shall be adjusted to pH 4.2 ± 0.1 by adding more or less of Solution A. The above stock Solutions A and B will keep in a stable condition indefinitely. Gum Arabic is added when making the press water prior to use. When Gum Arabic is added in mixing a stock solution, the shelf life is rather short.

5.1.4 Use.- The lithographic fountain solution is used in the dampening system of a lithographic press for moistening the plates and maintaining ink repellency of the non-image area of the plate throughout a printing run.

5.1.5 Safety precautions.- Inhalation of chromate salts and vapors may be injurious to health; skin contact may cause irritation. Contact with phosphoric acid solutions may burn the skin. Use with adequate ventilation and wear rubber gloves. If contacted accidentally, wash immediately with water. Personnel handling concentrated solutions should wear eye-protective equipment.

5.1.6 Packaging and labeling.- The fountain solution shall be packaged in quart or gallon polyethylene or polypropylene bottles. Type II fountain solution may be packaged in two solutions. Solution A can contain only the gum solution; solution B, the remainder of the formula. Solution A and solution B in separate containers shall constitute a unit. The label shall contain name, type of material, stock number, name of manufacturer, contract number. Containers containing dichromate salts (type II) shall include the following:

"POISON - CAUTION
AVOID INHALATION OF FUMES
USE WITH ADEQUATE VENTILATION
AVOID SKIN CONTACT
WASH THOROUGHLY WITH WATER FOLLOWING ACCIDENTAL SKIN CONTACT
DO NOT TAKE INTERNALLY"

and directions for preparation of the solution for use on the printing press.

5.1.7 Storage.- Keep container tightly closed when not in use.

5.2 Name.- Lithographic, blanket-roller, wash.

5.2.1 Specification.- O-L-298, Lithographic, Blanket-Roller, Wash (For Daily Wash).

5.2.2 Performance.- This solution surface cleans rubber blankets and rubber rollers.

5.2.3 Technical description.-

5.2.3.1 Type I.- Type I is a slow drying aliphatic petroleum distillate.

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TABLE I.- Chemical properties of the slow-drying blanket-roller wash

Characteristic	Temperature	Maximum percent	Minimum percent
Distillation	350°F. 375°F.	50.0 90.0	
End point °F. (maximum)	410°F.		
Distillation residue			1.5
Flash point (minimum)	100°F.		

5.2.3.1.1 The type I wash shall not contain any halogenated solvents, aromatic solvents, such as benzene, toluene, xylene, ketones or esters.

5.2.3.2 Type II.- Type II is a fast-drying mixture in the ratio specified in table II.

TABLE II.- Composition of fast-drying blanket-roller wash

Component	Percent by volume Minimum	Maximum
Drying cleaning solvent (P-D-680, type I)	59.0	61.0
Tetrachloroethylene (O-T-236)	24.0	26.0
Aliphatic petroleum naphtha	14.0	16.0

Aliphatic petroleum naphtha

Characteristic	Minimum	Maximum
Specific gravity at 60°F.	.730	.750
Distillation range:		
Initial boiling point (°F.)	240	---
50 percent distilled at (°F.)	---	260
End point (°F.)	---	290
Flash point, Tag Closed Cup (°F.)	55	---

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5.2.3.3 Physical properties.- Types I and II blanket-roller wash shall not leave any residual odor after drying. The flash point shall not be less than 100°F.

5.2.4 Use.-

5.2.4.1 General.- This blanket-roller wash is used for daily surface cleaning of rubber blankets and rubber rollers of offset duplicating presses. Neither type of wash is intended for immersion cleaning of the equipment.

5.2.4.2 Type I.- Type I slow-drying produces a minimum softening and swelling of rubber lithographic blankets.

5.2.4.3 Type II.- Type II is a fast-drying effective wash.

5.2.4.4 Directions for use.- Clean blanket and rollers with a wash-saturated cloth. Gentle wiping should remove the ink.

5.2.5 Safety precautions.- Flammable. Combustible. Keep away from heat, sparks and open flames. Vapors may be harmful. Use with adequate ventilation. May cause skin irritation. Dangerous if swallowed. Containers should be kept sealed when not in use.

5.2.6 Packaging and labeling.- The blanket roller wash is packaged in square or oblong base 1 gallon and 55 gallon drums. Containers must be labeled in accordance with Federal Hazardous Substances Labeling Act with the following:

LITHOGRAPHIC BLANKET-ROLLER WASH (FOR DAILY USE)
TYPE I OR TYPE II

Directions for use

Name of manufacturer

Contents

Contract number

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DIRECTIONS FOR USE

Use only for daily surface cleaning of rubber rollers of offset duplicating presses. Do not use for immersion cleaning of these items.

DANGER - POISON

FLAMMABLE - COMBUSTIBLE

"KEEP AWAY FROM HEAT, SPARKS AND OPEN FLAME
USE ONLY WITH ADEQUATE VENTILATION
AVOID PROLONGED OR REPEATED BREATHING OF VAPOR
KEEP CONTAINER SEALED WHEN NOT IN USE
AVOID PROLONGED OR REPEATED SKIN CONTACT
WASH THOROUGHLY FOLLOWING ACCIDENTAL SKIN CONTACT
DO NOT TAKE INTERNALLY
IF SWALLOWED DO NOT INDUCE VOMITING"

5.2.7 Storage.- Keep container sealed when not in use to prevent evaporation.

5.3 Name.- Surface plate coating solution.

5.3.1 Specification.- O-L-300, Lithographic Plate Sensitizing Coating, Surface.

5.3.2 Performance.- Surface press plates prepared with the sensitized coating solutions yield clear, legible impressions and no breakdown of image.

5.3.3 Technical description.- The coating solutions contain colloids dispersing agents, water and sensitizers. The manufacturer is given a wide latitude in the selection of component materials provided the resulting solutions are compatible with auxiliary materials used in the offset method of lithographic printing.

5.3.3.1 Class 1.- This is a two solution package consisting of a coating solution (A) and a sensitizing solution (B). The solution for use is prepared by diluting with water a mixture of four volumes of the coating solution (A) and one volume of sensitizing solution (B). The coating solution (A) shall be not darker than a light straw color.

5.3.3.2 Class 2.- Class 2 is a ready to use one solution package.

5.3.4 Use data.- These solutions are used for obtaining light-sensitive coatings on offset plates as used in the lithographic process of printing.

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5.3.5 Safety precautions.- Sensitizers containing chromates must be carefully handled. Inhalation of chromate salts and vapors may be injurious to health; skin contact may cause irritation. Use with adequate ventilation and wear rubber gloves. If contacted accidentally, wash immediately with water. Personnel handling concentrated solutions should wear eye protective equipment.

5.3.6 Packaging and labeling.-

5.3.6.1 Class 1, solution A.- Twenty-four ounces of solution A shall be put into a one-quart capacity (32 ounce), amber, wide-mouth type, Boston round bottle with a screw cap closure. The cap shall be fitted with suitable lining material which will not affect nor be effected by the product. A self-shrinking cellulose band shall be applied over the screw cap and extend down on the bottle to prevent loosening of metal cap while in transit. Alternatively, secondary closure may be effected by the use of pressure-sensitive plastic film sealing tape, three-fourths of an inch wide, applied over the screw cap and extending down on the bottle with tape overlapping one inch.

5.3.6.1.1 Class 1, solution B.- Six ounces of solution B shall be put into a six-ounce narrow-mouth bottle constructed and closed in accordance with the applicable requirements of PPP-C-186.

5.3.6.2 Class 2, sensitized.- One solution ready to use.

5.3.6.2.1 One-quart unit.- Thirty-two ounces of class 2 solution shall be put into a one-quart capacity amber bottle constructed and closed in accordance with the applicable requirements of PPP-C-186.

5.3.6.2.1.1 One-gallon unit.- One hundred and twenty-eight ounces of class 2 solution shall be put into a one-gallon-capacity amber bottle constructed and closed in accordance with the applicable requirements of PPP-C-186.

5.3.6.3 Labeling.- The labeling data shall be lithographed or legibly stenciled on each glass bottle with black weather-resistant and noncorrosive ink conforming to TT-I-558 or a combination consisting of a lithographed or stenciled label and a paper label may be used. The paper label shall be affixed to the bottle with a water insoluble, noncorrosive adhesive and coated overall with a transparent weather-resistant lacquer, enamel, or varnish shellac. The label data may also be furnished in the form of a decalcomania. The name of the product shall be stenciled or lithographed on the bottle body when paper labels are used. The directions for use information shall be not less than ten point type. Precautionary marking shall be not less than 12 point type in a conspicuous position. The label data shall be as follows:

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DIRECTIONS FOR PREPARATION

AND APPLICATION

A. Lithographic plate sensitizing coating, surface solution - unsensitized (two solutions).

For class 1

1. Preparation of sensitized solution. Pour solution B into bottle containing solution A. Mix thoroughly by shaking. To make a full batch of sensitized solution, dilute and mix thoroughly the above mixture A + B (30 ounces) with 11-1/4 ounces of water. If other quantities are desired, prepare solutions containing 73 parts of mixture A + B and 27 parts of water.

2. Application of sensitized solution. (Directions to be supplied by the manufacturer.)

B. Lithographic plate sensitizing coating, surface solution - sensitized (one solution - ready to use).

For class 2

1. Preparation of sensitized solution. Solution as received is ready to use without dilution.

2. Application of sensitized solution (Directions to be supplied by the manufacturer).

EXPIRATION DATE

KEEP COOL - PROTECT FROM FREEZING

Note: Any solution containing dichromate salt shall include on the label:

" CONTAINS DICHROMATE SALT
POISON - CAUTION
AVOID INHALATION OF FUMES
USE WITH ADEQUATE VENTILATION
AVOID SKIN CONTACT
WASH THOROUGHLY WITH WATER FOLLOWING ACCIDENTAL SKIN CONTACT
DO NOT TAKE INTERNALLY "

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5.3.7 Storage, class 1.- The combined coating and sensitizing solutions shall be stored in tightly stoppered glass bottles for a period of 7 days at a temperature between 40° to 50°F. The solutions shall be examined for clarity immediately after mixing and after the 7 day period.

5.4 Name.- Lithographic, solvent.

5.4.1 Specification.- O-L-1031, Lithographic Solvent.

5.4.2 Performance.- Lithographic solvent removes dried and oxidized ink deposits and other stubborn soils from the press.

5.4.3 Technical description.-

5.4.3.1 Composition.- The solvent shall have the following composition:

TABLE III

Component	Percent by weight
Pine oil	9.4 ± 1.0
Castor oil	1.0 ± 0.1
Ester gum	2.0 ± 0.1
Dry cleaning solvent	87.6 ± 2.0

5.4.3.2 Physical properties.- The solvent has a pleasant pine odor and is no darker than straw in color. The flash point shall be not less than 100°F.

5.4.4 Uses and limitations.-

5.4.4.1 General.- The solvent is used to remove dried and oxidized ink and stubborn soils from lithographic presses. It is not to be used on rubber offset blankets or rollers because it may swell the rubber causing softening of the roller and delamination of the blanket.

5.4.4.2 Directions for use.- Remove dried and oxidized ink deposits and other stubborn soils from the press by rubbing the soiled parts with a pad of clean cloth dampened with the solvent cleaner.

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5.4.5 Safety precautions.- Prolonged or repeated skin contact may cause irritation. Should not be taken internally, but if accidentally swallowed, do not induce vomiting. Seek medical attention immediately.

5.4.6 Packaging and labeling.- This solution is supplied in one gallon metal cans. Labeling shall be in accordance with the Federal Hazardous Substances Labeling Act with the following information:

DIRECTIONS FOR USE

Remove dried and oxidized ink deposits and other stubborn soils from the press by rubbing the soiled parts with a pad of clean cloth dampened with solvent cleaner. Do not use this cleaner on rubber offset blankets or rollers because it may swell the rubber causing softening of the roller and delamination of the blanket.

" DANGER - FLAMMABLE
KEEP AWAY FROM HEAT, SPARKS AND OPEN FLAME
USE WITH ADEQUATE VENTILATION
AVOID INHALATION OF VAPORS
AVOID PROLONGED OR REPEATED SKIN CONTACT
DO NOT TAKE INTERNALLY
IF SWALLOWED, DO NOT INDUCE VOMITING "

5.4.7 Storage.- Container should be kept tightly closed when not in use.

5.5 Name.- Regenerator, lithographic blanket.

5.5.1 Specification.- O-R-1151, Regenerator, Lithographic Blanket.

5.5.2 Performance characteristics.- The blanket regenerator shall temporarily swell the blanket to correct low areas in the blanket.

5.5.3 Technical description.- The blanket regenerator is a semi-fluid or paste consisting of a mixture of organic solvents, thickening agents, and wetting agents.

5.5.4 Use data.-

5.5.4.1 General.- The blanket regenerator is for extending the service life of rubber blankets of offset duplicating presses by temporarily swelling the rubber and correcting low areas of blankets caused by uneven wear, tension on the blanket or smashup.

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5.5.4.2 Directions for use.- Apply, by brush, a layer of the blanket regenerator to the low area of the blanket. Allow regenerator to remain in contact with blanket for about 1 minute. Wipe off regenerator using soft cloth dampened with a lithographic blanket-roller wash. If low spot on blanket appears after a few impressions, repeat the above treatment with regenerator. On some type blankets, the regenerator should remain on the blanket for more or less time.

5.5.5 Safety precautions.- Flammable. Combustible. Keep away from heat, sparks and open flames. Vapors are harmful. Use only with adequate ventilation. May cause skin irritation. Dangerous if swallowed. Container should be kept sealed when not in use.

5.5.6 Packaging and labeling.- The regenerator shall be packaged in a 4 fluid ounce glass bottle. Each bottle shall be provided with a synthetic brush commercially used for the specific product being packaged and the brush shall be an integral part of the cap. Labeling shall be in accordance with the Federal Hazardous Substances Labeling Act with the following information in 10 point type:

DANGER - CONTENTS VOLATILE AND COMBUSTIBLE
AVOID INHALATION OF FUMES
PROVIDE ADEQUATE VENTILATION IN WORK AREAS
HARMFUL OR FATAL IF SWALLOWED
AVOID PROLONGED OR REPEATED CONTACT WITH SKIN
IF SWALLOWED, DO NOT INDUCE VOMITING
KEEP CONTAINER SEALED WHEN NOT IN USE

Each unit container shall also be labeled as follows:

DIRECTIONS FOR USE

Apply, by brush, a layer of the blanket regenerator to the low area of the blanket. Allow regenerator to remain in contact with blanket for about 1 minute. Wipe off regenerator using a soft cloth dampened with a lithographic blanket-roller wash. If low spot on blanket appears after a few impressions, repeat the above treatment with regenerator. On some type blankets, the regenerator should remain on the blanket for more or less time.

5.5.7 Storage.- Highly volatile. Keep bottle tightly closed when not in use.

5.6 Name.- Lithographic, plate, asphaltum solution.

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5.6.1 Specification.- TT-L-132, Lithographic Plate Asphaltum Solution.

5.6.2 Performance.- Lithographic plate, asphaltum solution keeps the image on the offset duplicating plates ink receptive.

5.6.3 Technical description.- Lithographic plate asphaltum solution consists of asphaltum dissolved in turpentine or volatile mineral spirits.

5.6.4 Use data.-

5.6.4.1 General.- The lithographic plate asphaltum solution is intended for use as a protective coating for images on offset duplicating plates and to make the image areas more ink receptive. The material is to be applied after developing ink has been removed with a solvent.

5.6.5 Safety precautions.- This is a flammable and toxic material. It should be kept away from heat, sparks and open flame. Use in well ventilated areas.

5.6.6 Packaging and labeling.- The asphaltum solution is packaged in 1 quart and 1 gallon cans. The label shall be in accordance with the Federal Hazardous Substances Labeling Act with the following information:

DANGER: CONTENTS VOLATILE AND COMBUSTIBLE
AVOID INHALATION OF FUMES
PROVIDE ADEQUATE VENTILATION IN WORK AREAS
HARMFUL OR FATAL IF SWALLOWED
AVOID PROLONGED OR REPEATED CONTACT WITH SKIN
PROLONGED OR REPEATED SKIN CONTACT MAY CAUSE SKIN IRRITATION

Warning! Flammable. Keep away from heat, sparks and open flame.

5.6.7 Storage.- Highly volatile. Keep bottle tightly closed when not in use.

5.7 Name.- Lithographic plate preserving gum.

5.7.1 Specification .- JJJ-L-821, Lithographic Plate Preserving Gum.

5.7.2 Performance.- The Lithographic preserving gum is used for protection against marking, abrasion, grease, and oxidation during storage.

5.7.3 Technical description.- A gum arabic is used to prepare the lithographic plate preserving gum solution. It shall contain a mold inhibitor which shall keep the gum solution in condition for use for a period of one to one and one-half years.

5.7.3.1 Physical properties.- The gum solution shall have a specific gravity of 1.058 $\pm$ 0.005 at 15.56°/15.56°C. or a Baumé reading of 8° $\pm$ 0.5° at 60°/60°F.

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5.7.4 Use data.- The preserving gum is used on press plates for protection against marking, abrasion, grease and oxidation during storage.

5.7.4.1 Directions for use.- With a slightly dampened sponge, spread the gum over the test plate in both directions of the plate. Rub down smoothly with cheesecloth so that a thin, even coating of gum covers the plate. Fan the plate dry.

5.7.5 Packaging.- The solution is packaged in 1 quart or 1 gallon polyethylene bottles.

5.7.6 Storage.- The shelf life is for a period of one to one and one half years.

Custodians:

Army - GL
Navy - SH
Air Force - 68

Preparing activity:

Army - GL
Project No. 6850-0284

Code C

Review activities:

Army - GL, MD, MU
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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